

Networks and Systems: MM1

Hans-Peter Schwefel

- **Mm1** Introduction to Systems and Networks (hps)
- **Mm2** Layer-2 Example Technologies: Field-buses CAN and Flexray (hps)
- **Mm3** TCP Modeling and Performance Analysis (hps)
- **Mm4** Network Calculus I (HS)
- **Mm5** Network Calculus II (HS)

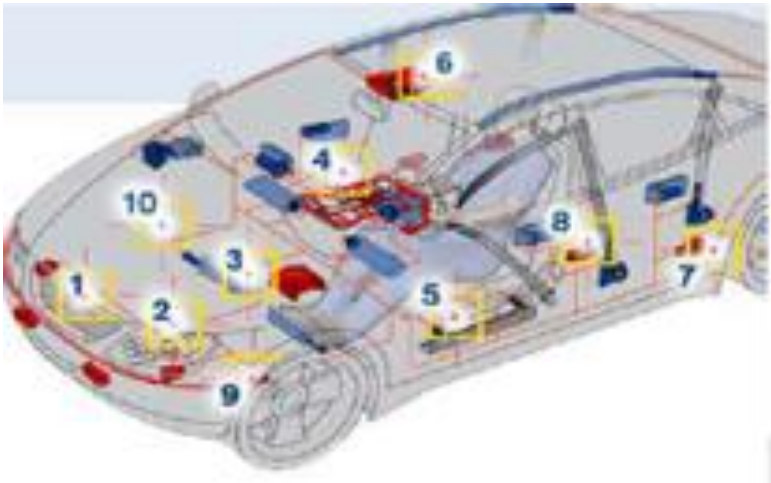
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Networks and Systems - Motivation

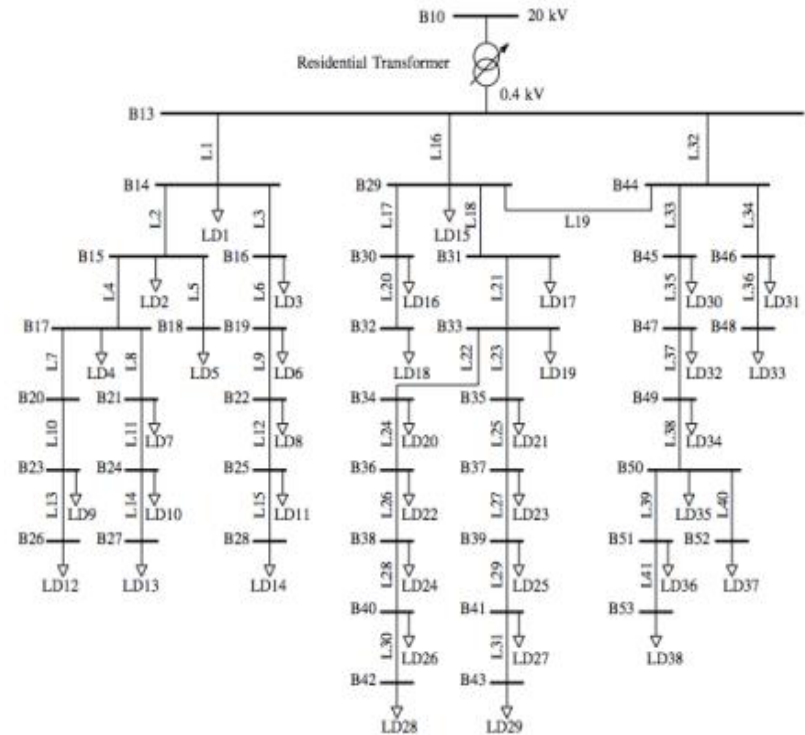
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Rational:



Example systems

- Automotive:
 - ▣ In-vehicle control systems (engine, breaking, ...)
 - ▣ autonomic driving using car-2-car or car-2-infrastructure communication
- Intelligent Electricity Grids



System definition

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System (Wikipedia): A **system** is

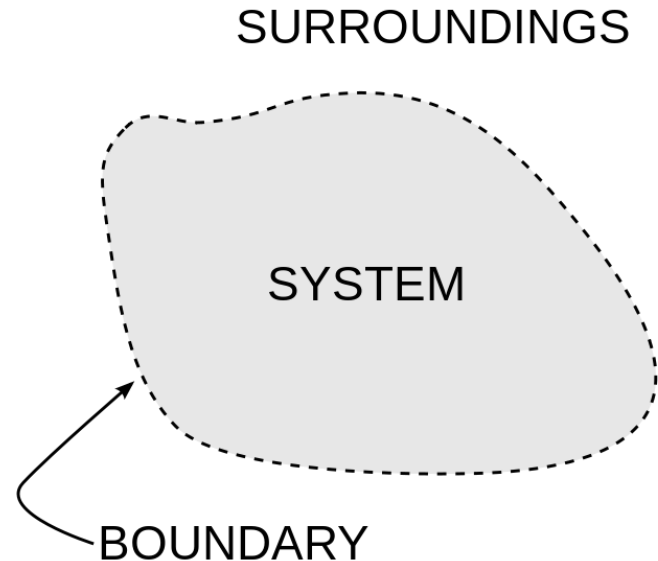
- ▣ a group of interacting or interrelated elements
- ▣ that act according to a set of rules to form a unified whole

Categories that you have previously seen

- ▣ Distributed System (Comtek-5)
 - Important 'tool': space-time diagrams (message sequence chart)
- ▣ Control System: Physical System, Sensors, 'Upstream' Network, Controller, 'Downstream' Network, Actuators

System scope: boundaries

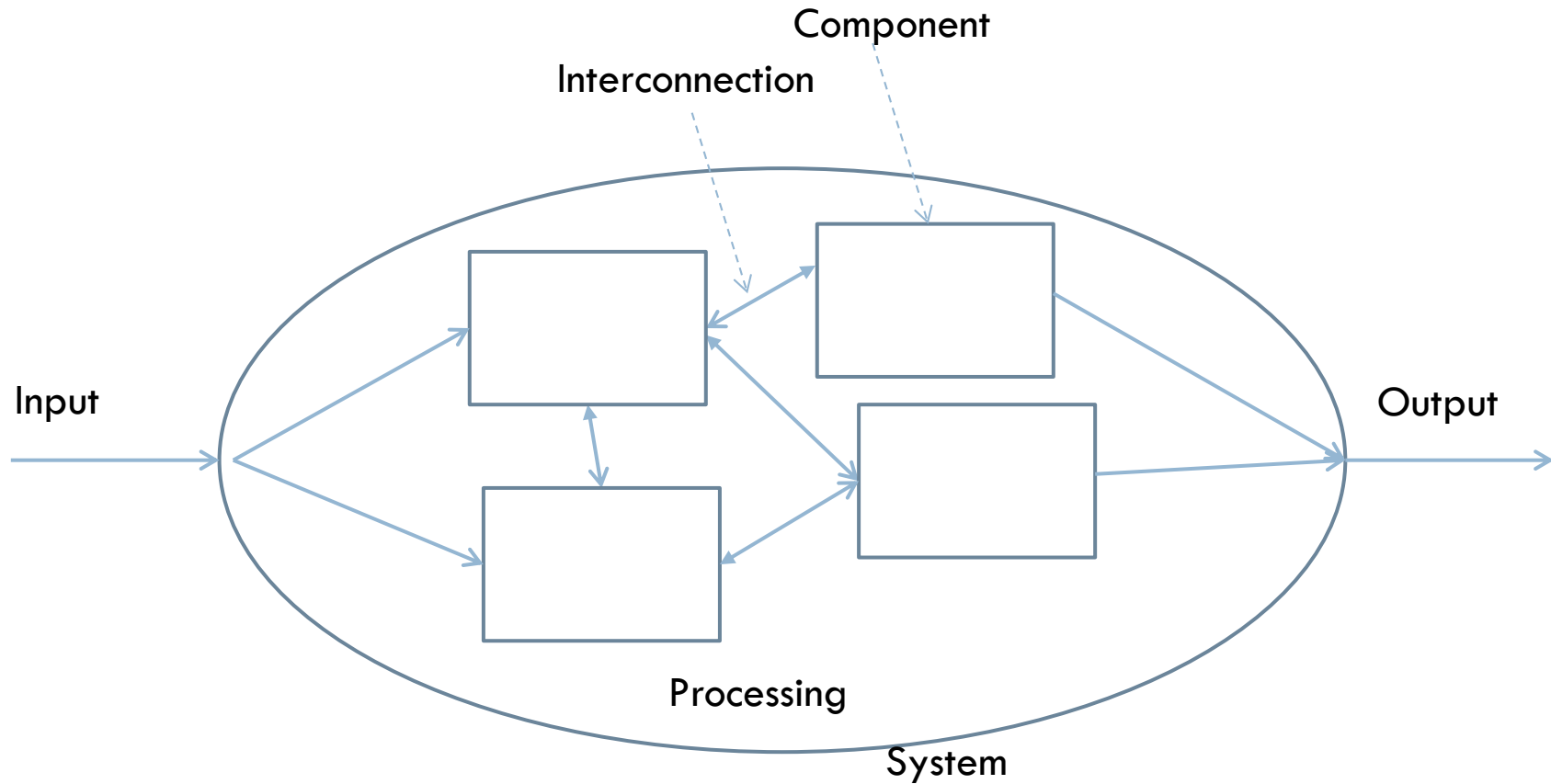
- We scope a system by defining its **boundary**; this means choosing which entities are **inside** the system and which are **outside**.



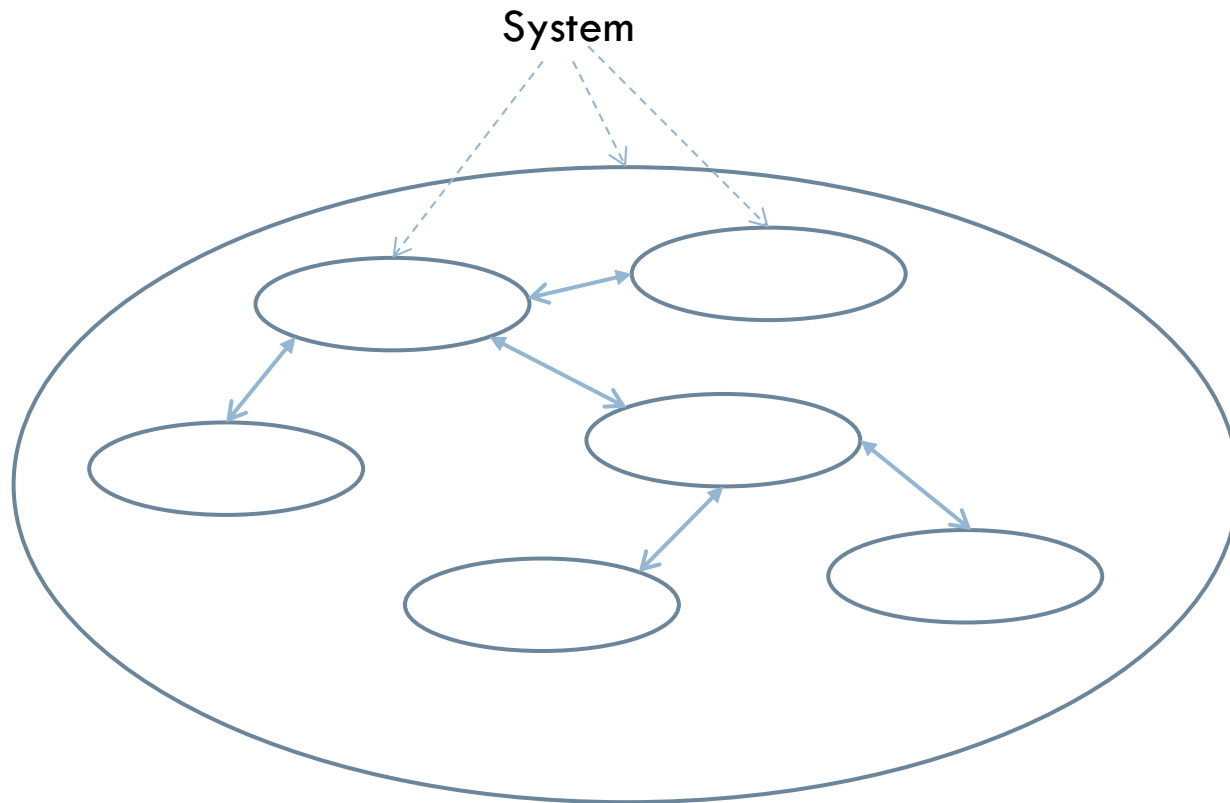
System

- Systems have **structure**, defined by components/elements and their composition;
- Systems have **behavior**, which involves inputs, processing and outputs of material, energy, information, or data;
- Systems have **interconnectivity**: the various parts of a system have functional as well as structural relationships to each other.

System structure



System of Systems



Components are systems in their own right.
Each subsystem serves an individual purpose.

System of Systems - principles

What characterizes a system of systems:

- Operational Independence of subsystems
- Managerial Independence of subsystems
- Evolutionary Development (and deployment)
- Emergent Behavior
- (Geographical) Distribution of Subsystems

Source: Maier, M.W. (1998). "Architecting Principles for System of Systems". Systems Engineering. 1 (4): 267–284.

Others sometimes demanded in 'addition' (DeLaurentis, D. "Understanding Transportation as a System of Systems Design Problem," 43rd AIAA Aerospace Sciences Meeting, Reno, Nevada, January 10–13, 2005):

- Interdisciplinarity, Heterogeneity of components, Networks of Systems

Principles - Details

M.Maier 1998

- **Operational Independence** of the Elements: If the system-of-systems is disassembled into its component systems the component systems must be able **to usefully operate independently**. The system-of-systems is composed of systems which are independent and **useful in their own right**.
- **Managerial Independence** of the Elements: The component systems **not only can** operate independently, **they do operate independently**. The component systems are separately acquired and integrated but maintain a continuing operational existence independent of the system-of- systems.
- **Evolutionary Development**: The system-of-systems does not appear fully formed. Its development and existence is evolutionary with functions and purposes added, removed, and modified with experience.
- **Emergent Behavior**: The system performs functions and carries out purposes that do not reside in any component system. These behaviors are emergent properties of the entire system-of-systems and cannot be localized to any component system. The principal purposes of the systems-of-systems are fulfilled by these behaviors.
- **Geographic Distribution**: The geographic extent of the component systems is large. Large is a nebulous and relative concept as communication capabilities increase, but at a minimum it means that the components **can readily exchange only information and not substantial quantities of mass or energy**.

Example: Wind-farm controller

- Central controller for multiple wind-turbines with targets

- Achieve reference generation
- Reduce fatigue – accumulated damage

- controller communicates with wind turbines over Access Network

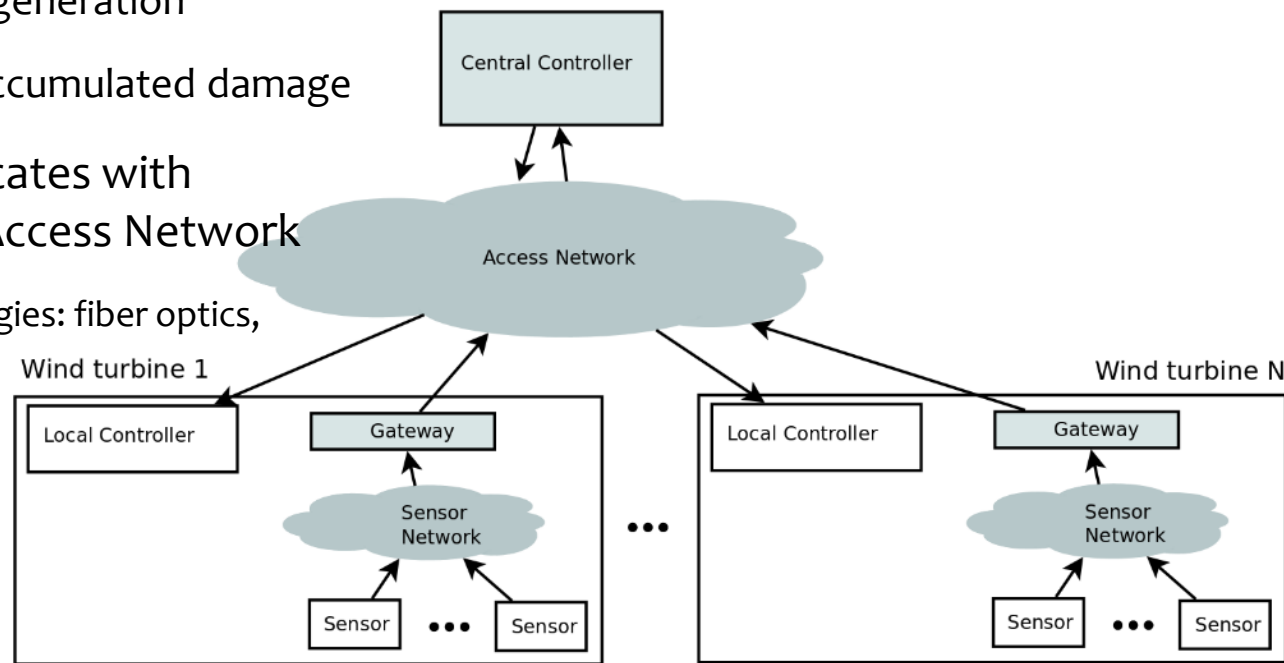
- Examples of technologies: fiber optics, LTE, 3G, 2G, PLC

- Local controller acts on set-points from central controller

- Sensors periodically provide measurements over Sensor Network to gateway

- Examples of technologies: Fieldbus, CANbus, WSN

- 3 sensors on each wind turbine: Torque x 2, angle rotation



Discussion: Wind-farm control system

- Operational Independence of subsystems ?
- Managerial Independence of subsystems ?
- Evolutionary Development ?
- Emergent Behavior ?
- (Geographical) Distribution of Subsystems ?

‘Network’ impact example:

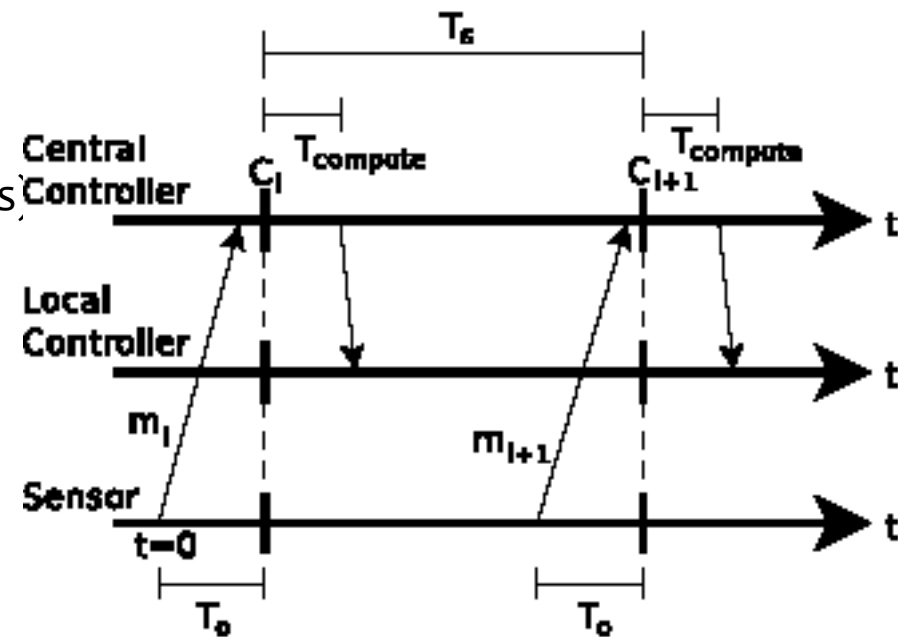
Sensors send at deterministic periodic with configurable offset

- Simplified message sequence diagram

- T_s : Control period (150 ms)
- $T_{compute}$: computation time (50 ms)
- T_o : offset
- C_i : Computation instant

- Higher T_o gives better chance of reaching C_i in time,

- however increasing ‘age’ of the information



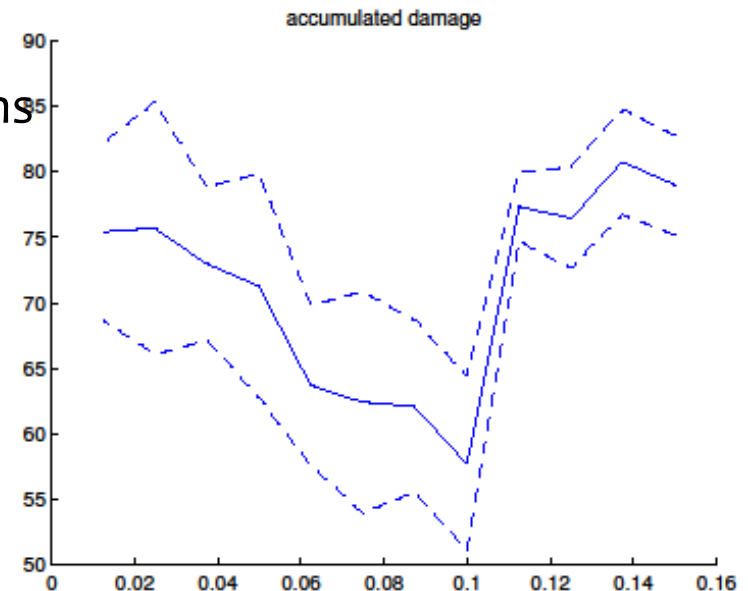
Controller performance results: impact of varying offsets

Scenario:

- 30 ms mean exponentially distributed upstream delays, shifted 1 ms
 - Neglect downstream delays for now (0ms)
- Vary offset between 10ms and 150ms

Observations:

- Accumulated damage decrease as T_o increases, until $T_o = 100$ ms



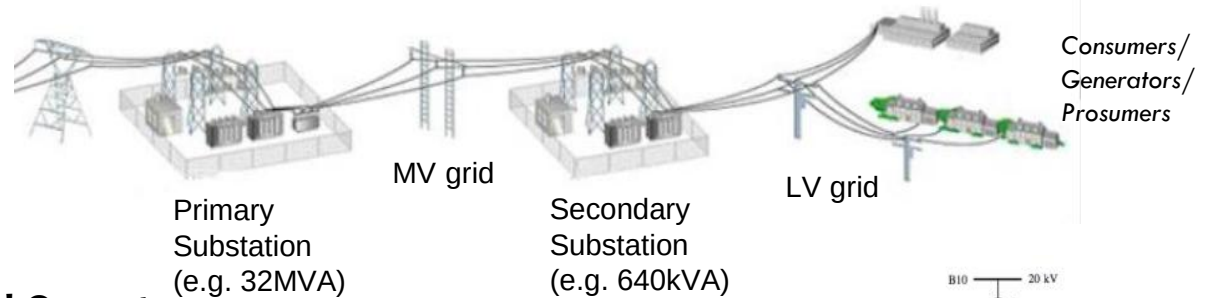
Explanation: At $T_o = 100$ ms wind turbine receives previous controller set-point

- This means that at higher $T_o > 100$ ms sampling from the system with the old set-point is used for the next control computation

Example: Electricity Distribution Grids

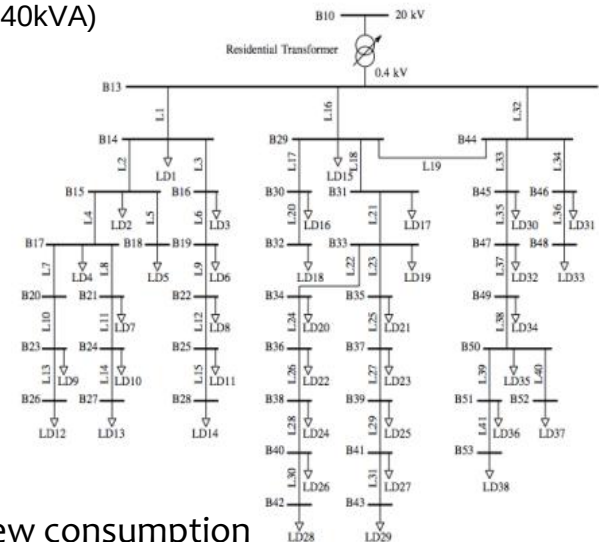
Hierarchical electricity grid structure

- Medium Voltage (10kV, 20kV or 30kV)
- Low Voltage (400V)



Targets of Distribution Grid Operator

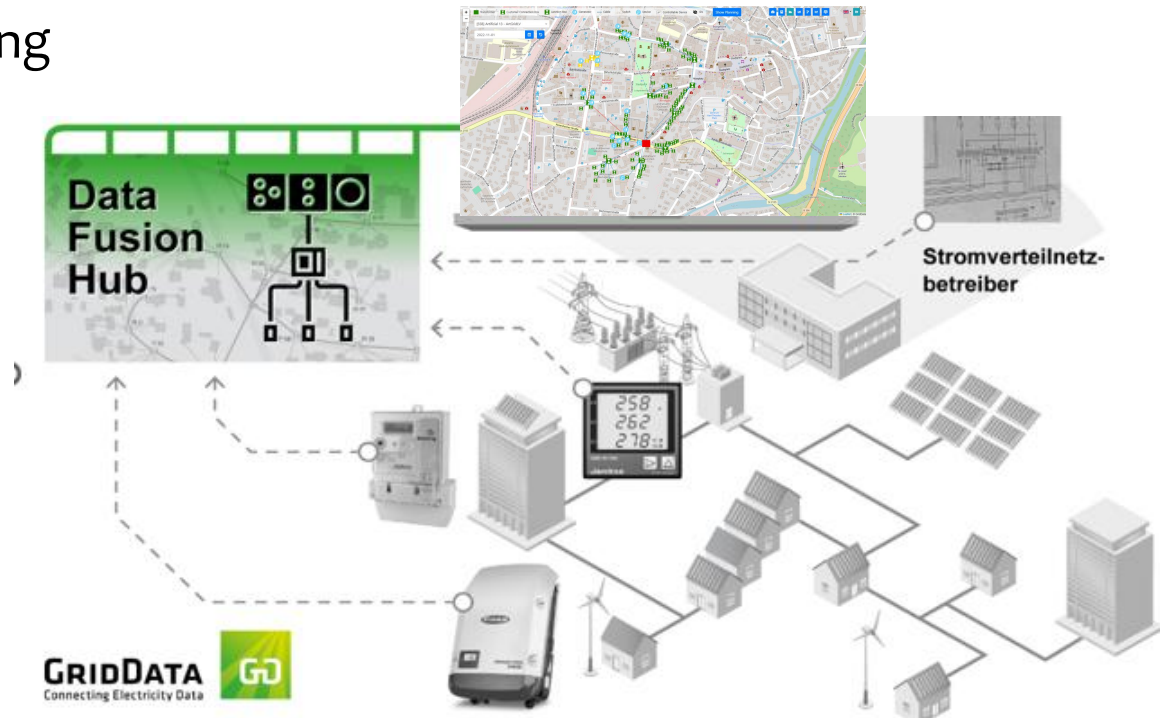
- Reliable distribution of electrical energy
minimize outage durations & impact, avoid overload and faults
- Voltage quality
Assure voltage within $\pm 10\%$ (or $\pm 6\%$) of nominal voltage
- Efficiency
Minimize energy losses in the grid
Increase life-time of grid components
Target grid investments for renewable generation integration and new consumption



Example: Intelligent Electricity Distribution Grids

e.g. SCADA systems (Medium and High Voltage mainly) or more advanced Digital Twins:

- Observation by measurement devices
- (Automated) Model Building from grid structural information
- Used for monitoring, control, and planning
 - ▣ Representing/based on the true electrical grid behavior

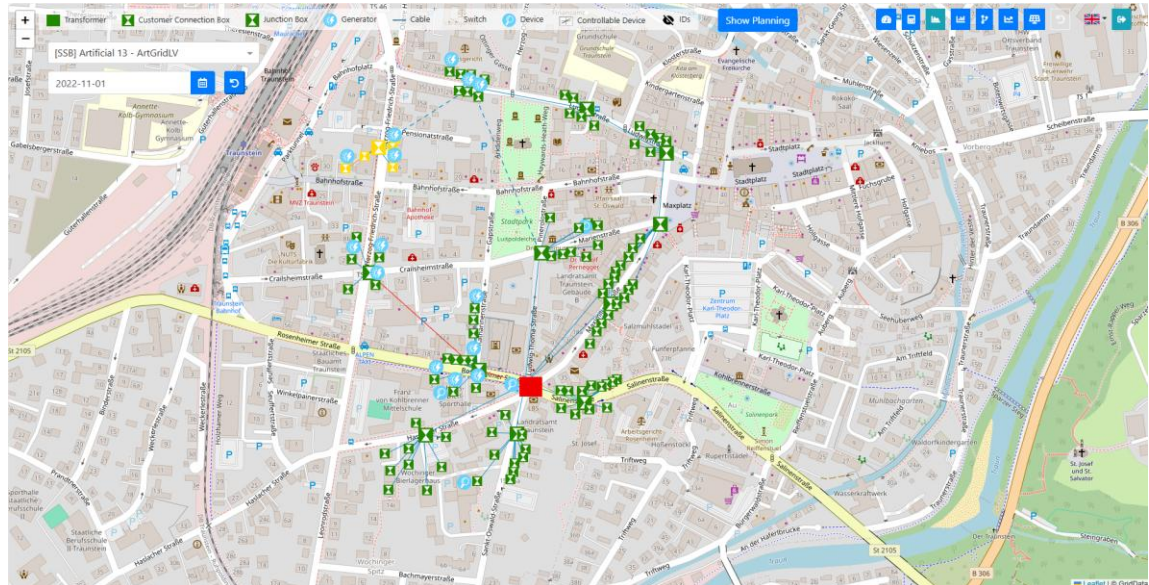


Example: Intelligent Electricity Distribution Grids

Subsystems: Physical Grid Components (Cables, Trafos); Measurement Devices; Communication Networks to access measurements; IT Systems to store measurements; remotely controllable (configurable) components such as inverters, breakers, trafos, flexible consumers, ...

□ System boundaries, e.g. include?

- Customer behavior (consumption, generation)
 - Boundary at Smart Meter?
- Electricity markets (kWh prices)
- Higher-layer grid behavior



Discussion: Intelligent electricity distribution grids

- Operational Independence of subsystems?
- Managerial Independence of subsystems ?
- Evolutionary Development ?
- Emergent Behavior ?
- (Geographical) Distribution of Subsystems ?

→ Detailed Scope for Networks & Systems Lecture: How does sensor data access (data collection, communication network, Intermediate data processing and storage, ...) influence properties of monitoring and control applications?

What does this imply for your semester project

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Systems of Systems

- Operational Independence of subsystems
- Managerial Independence of subsystems
- Evolutionary Development
- Emergent Behavior
- (Geographical) Distribution of Subsystems

Plus:

- Interdisciplinarity, Heterogeneity of components, Networks of Systems

→ Recommended to at least be aware of Systems of Systems aspects of your project system for potentially enlarged system boundaries

‘Relaxed’ derived criteria for **semester projects**: include several of

- Sensing of some physical system through **multiple distributed sensor devices**
- **Communication** of the sensing units **via a network** towards one or more processing units that infer the state of the sensed system (in a centralized or distributed manner)
- **Computation** implementing some (centralized or distributed) optimization over the inferred state
- Feedback to actuators in the physical system for the purpose of **control**
- Use of **heterogeneous sensors**
- **Modeling and assessment** of the behavior of the system for different assumptions on the underlying information exchange (reliability, periodicity, delay, sparsity, failure mechanisms, data integrity, etc.)

Networks and Systems Course Goal

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- Understand the design choices and performance behavior of
 - selected practical technologies (IP, field buses, ...)
 - and selected use-cases including scenarios of faultsfor access to sensor data in networked systems
- Be able to apply different modeling and assessment techniques to analyze networked systems
 - Focus (mostly) on discrete event systems

Content

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1) System definitions and examples

- Systems, Systems of Systems, Semester Project scope
- Examples: Wind-farm controller, intelligent distribution grids

2) End-2-end Network Performance

- OSI layer and related performance metrics
- Layer-3 end-to-end performance - delay, loss, throughput

3) Stochastic discrete event models: Markov chains and their application to Layer 2 and Layer 3 performance – Buffering and multiplexing in intermediate nodes

- Continuous Markov models and birth-death processes (revision)
 - M/M/1 queueing models
 - More general models and network traffic characterization
- See C&L Sect 7.3/7.4, 8.1-8.6*

Outlook Next lectures

Layer-2 Examples: Field-buses CAN & Flexray

Layer 4 - TCP performance models

Performance assessment of data access schemes

OSI Layers revision

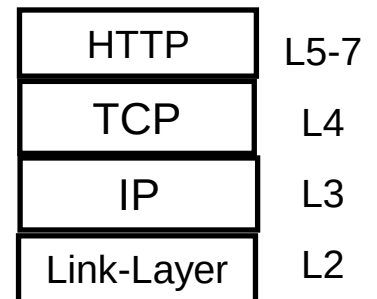
- Application Layer (L7)
 - Application layer protocols HTTP(s), FTP(s), SSH, ...
 - Used by applications
 - Applications web-browsing, file-transfer, remote login, VoIP, Video Conference, Gaming, ...
- Presentation Layer (L6)
 - e.g. html, xml, JSON, audio codec, etc.
- Session Layer (L5)
 - E.g., SIP (Session Initiation Protocol)
- Transport Layer (L4)
 - TCP (Transmission Control Protocol), TLS, UDP (or others SCTP,...)
- Network Layer (L3)
 - E.g. Internet Protocol (IP)
- Link-Layer (L2) + Physical Layer (L1)
 - E.g. Ethernet, WLAN 802.11, Bluetooth, CAN, Flexray,...

HTTP	L5-7
TCP	L4
IP	L3
Link-Layer	L2

Remark: Cellular technologies 5G, 4G, 3G, 2G themselves contain a more complex internal 'network' structure, containing internally different protocol layers covering at least Layer 1-4; however, they can be seen as access technology to an IP network, then they take over the role of Layer-1 and 2

OSI Layers revision

- Physical Layer (L1) – transmission of symbols over medium
- Link-Layer (L2) – transmission of L2 frames on a medium
 - Addressing, FEC/CRC, Medium Access Control, and Logical Link Control
- Network Layer (L3) – transmission of packets end-to-end between hosts
 - transmission between end-nodes (hosts) over a network with intermediate hops (routers)
 - routing
 - L3 addressing
- Transport Layer (L4) – end-to-end communication between processes on hosts
 - Process addressing on host via port numbers, Concept of L4 addresses (IP address + port number)
- Session Layer (L5)
- Often implicit, e.g. session established via cookies or tokens in web browsing sessions
- Presentation Layer (L6), e.g. html format, JSON, audio codec, etc.
 - representation of data
- Application Layer (L7), e.g. HTTP(s), FTP(s), SSH, VoIP, Video Conference, Gaming, ...



Supporting protocols, selected examples

- ARP – Address Resolution protocol – Linking Layer 3 address to Layer 2 addresses
- DNS – Domain Name Service – translating hierarchical names to IP addresses (or address ranges)

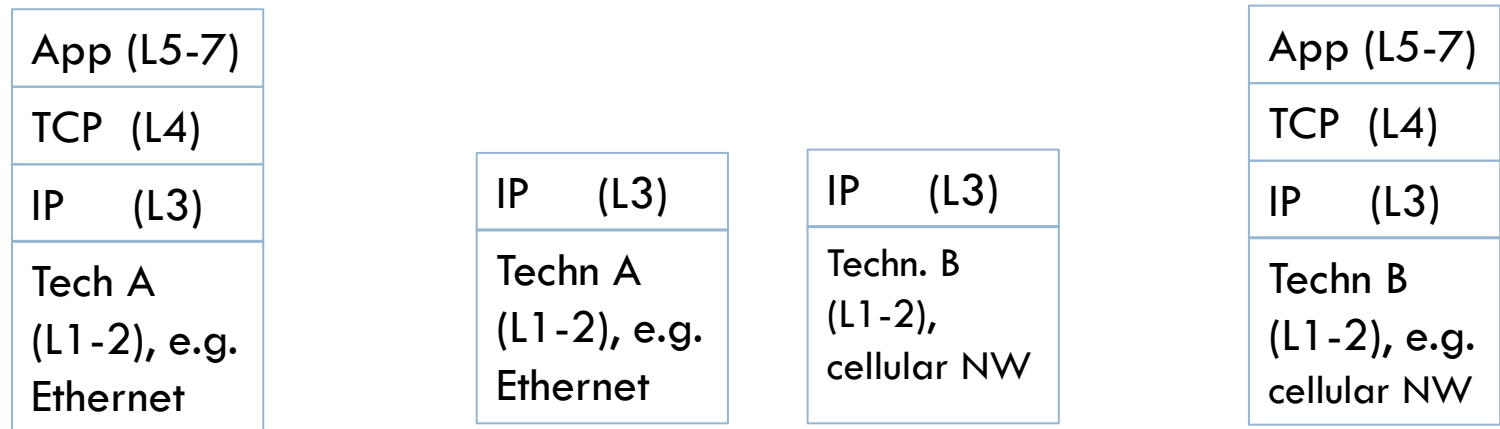
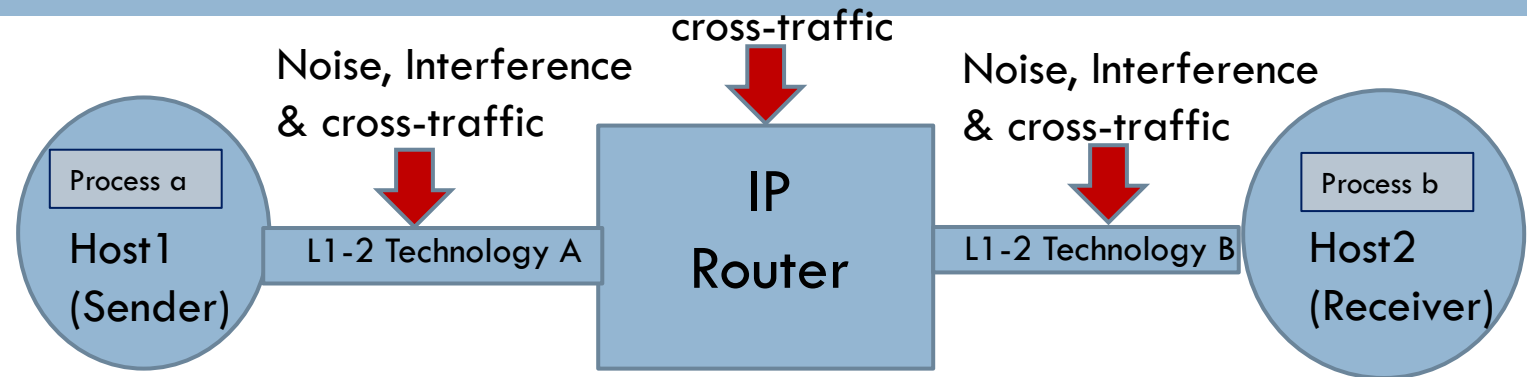
OSI Layers: Data units and performance metrics

- Physical Layer (L1): Symbol
 - Symbol transmission duration
 - Symbol decoding success probability
- Link-Layer (L2): L2-Frame
 - Frame Delay (one way or round-trip)
 - Frame Transmission Success probability,
 - L2 Throughput, L2 Goodput
- Network Layer (L3): IP Packet
 - End-to-end IP Packet Delay
 - Packet Delivery Probability = 1- Packet Loss Probability
 - L3 Throughput, L3 Goodput (Layer-3 Payload per time unit)
- Transport Layer TCP (L4): TCP Segment
 - End-to-end delay, round-trip time delay
 - L4 throughput, L4 Goodput
 - Other metrics may depend on protocol used, in particular reliability metrics
- Application Layer: application-specific messages or data streams
 - Application specific metrics, e.g. in wind-farm example
 - rereference tracking accuracy, accumulated damage
 - Application protocol behavior: message delays, reliability, throughput/goodput

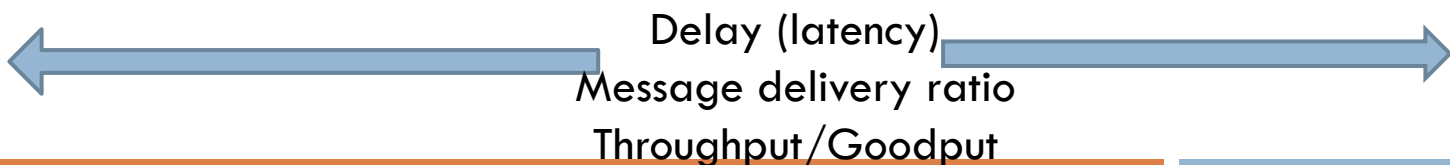
HTTP	L5-7
TCP	L4
IP	L3
Link-Layer	L2

Starting with a small network: 2-hop setting with generic link-layer

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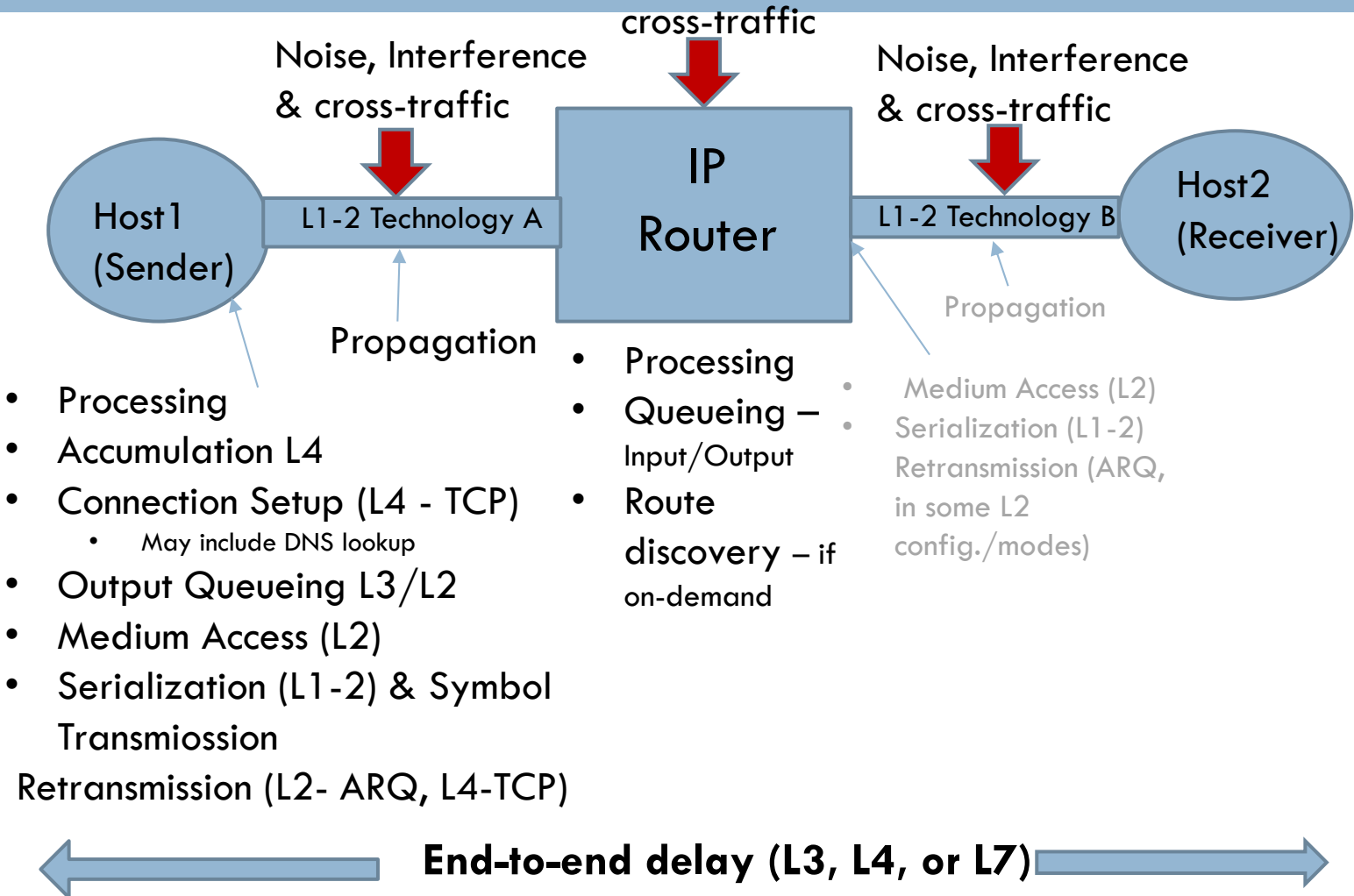


End-to-end metrics (L3, L4, or L7):



Delay (latency)

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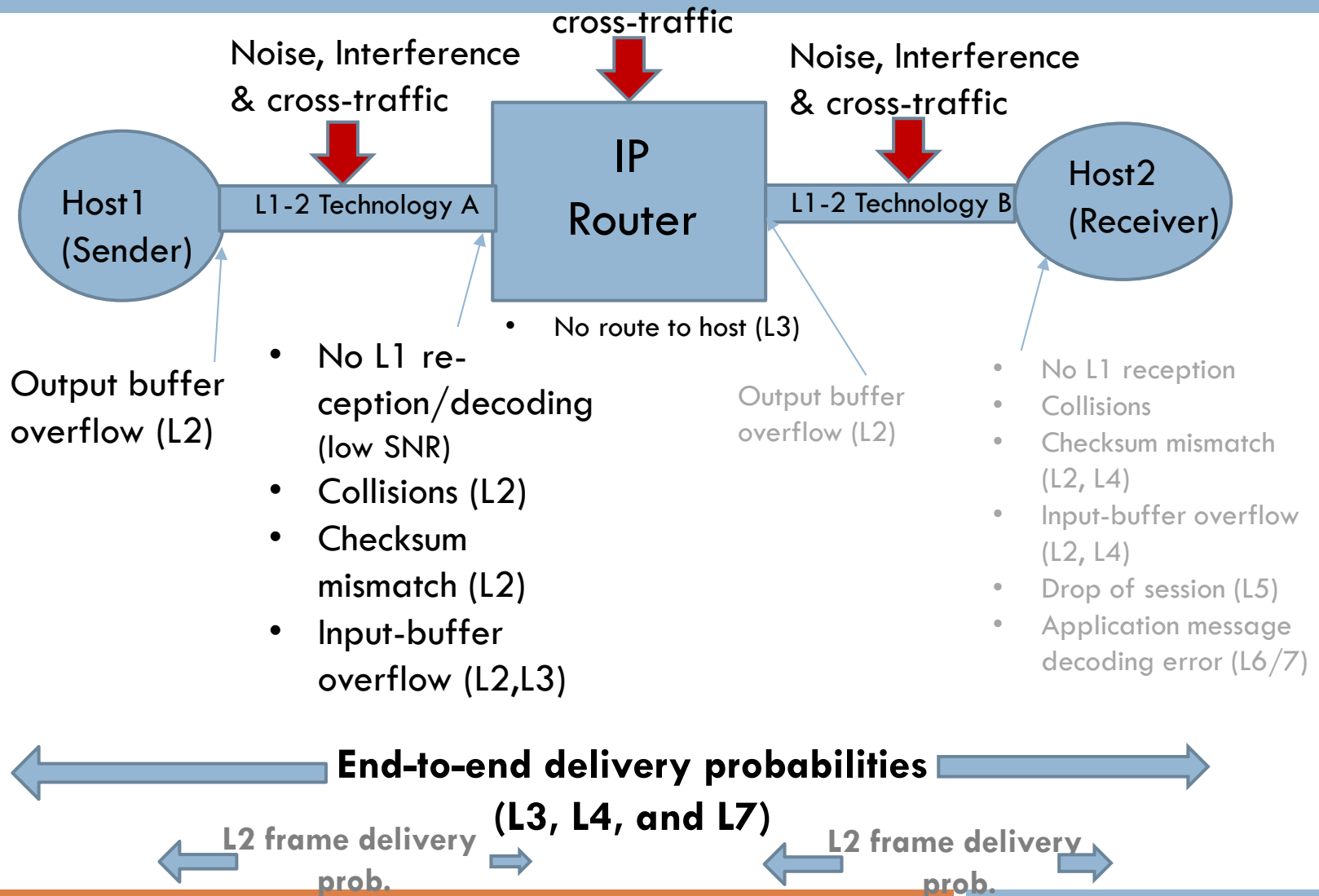
Exercise

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- Assume a Layer-2 technology of your choice (or the one used in your project), e.g. WLAN, Bluetooth, Ethernet
 - What are the orders of magnitude of the different delays on the previous slide?
 - What dependencies are there, e.g. which delay depends on payload size, interference, etc.; which of these dependencies are candidates for representation with stochastic distributions, what distribution types?

Losses

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Exercise: Throughput

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- Given a L1 technology with symbol rate $10^6/\text{s}$ and 4 bits per symbol. Layer-2 uses a 12 byte header and 80 byte payload and 8 byte CRC code.
 - ▣ Calculate the maximum theoretical L2 payload throughput when neglecting MAC delays and assuming no losses
 - ▣ Under which scenarios will propagation delays impact throughput?
 - ▣ Now assume a 10% frame loss probability and a medium access delay of initially 10 microseconds. Upon each retransmission, the medium access delay is doubled. Calculate the throughput for this case.
 - ▣ The second link uses a L2 technology with same symbol type, same symbol rate and same frame structure but now 100 byte payload. What is the theoretical achievable L2-throughput via the 2 links in series assuming ideal conditions (no MAC delay, no frame losses).

Content

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- Continuous Markov models and birth-death processes (revision)
- M/M/1 queueing models
- More general models and network traffic characterization

See C&L Sect 7.3/7.4, 8.1-8.6

Outlook Next lectures

Layer-2 Examples: Field-buses CAN & Flexray

Layer 4 - TCP performance models

Performance assessment of data access schemes

System Models – Categories and lecture scope

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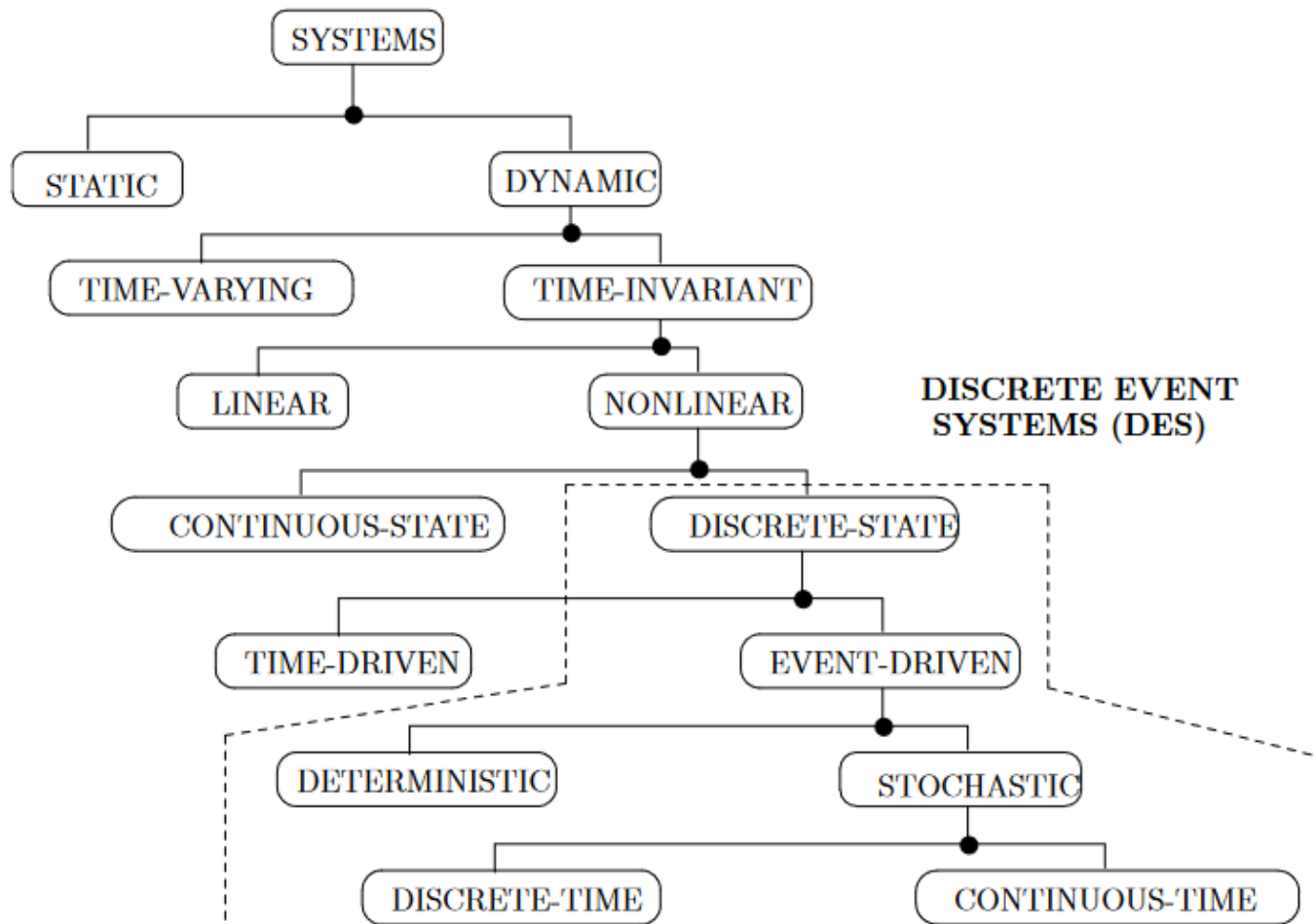


Figure 1.31: Major system classifications.

Source: C&L Sect. 1.4

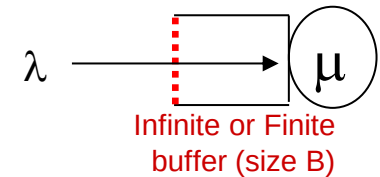
Continuous Time Markov Processes

- Defined by
 - ▣ **State-Space:** finite or countable infinite, w/o.l.g. $E=\{0,1,2,\dots,K\}$ ($K=\infty$ also allowed)
 - ▣ **Transition rates:** μ_{jk}
 - Holding time in state j : exponential with rate $\sum_{k \neq j} \mu_{jk}$
 - Transition probability from state j to k : $\mu_{jk} / \sum_{l \neq j} \mu_{jl} =: \mu_{jk} / \mu_j$
- X_t = RV indicating the current state at time t ; $\pi_i(t) := \Pr(X_t = i)$
- 'Markov Property': transitions do not depend on history but only on current state
 $t_0 < t_1 < \dots < t_n, \forall i_0, i_1, \dots, i_n, j \in E$
$$\begin{aligned} P\{X(t_{n+1}) = j \mid X(t_n) = i_n, \dots, X(t_0) = i_0\} = \\ P\{X(t_{n+1}) = j \mid X(t_n) = i_n\}, \end{aligned}$$
- Computation of transient and steady-state probabilities
 - ▣ Chapman Kolmogorov Equations: $d\pi_i(t)/dt = -\mu_i \pi_i(t) + \sum_{j \neq i} \mu_{ji} \pi_j(t)$
 - ▣ Flow-balance equations, steady-state ($t \rightarrow \infty$): $\mu_i \pi_i = \sum_{j \neq i} \mu_{ji} \pi_j$

Queueing Models: Kendall Notation

$X/Y/C[/B]$ Queues (example: $M/M/1$, $GI/M/2/10$, $M/M/10/10$, ...)

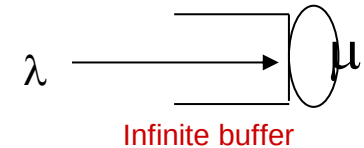
- **X: Specifies Arrival Process**
 - ▣ M =Markovian $\rightarrow$ Poisson (i.i.d. Exponential inter-arrival times)
 - ▣ GI =General Independent $\rightarrow$ iid
- **Y: Specifies Service Process ($M, G(I), \dots$)**
- **C: Number of Servers**
- **B: size of finite waiting room (buffer)**
[also counting the packet in service]
 - ▣ If not specified: $B = \infty$
- **Often also specified: service discipline**
 - ▣ FIFO: First-In-First-Out (default)
 - ▣ Processor Sharing: PS
 - ▣ Last-in-first-out LIFO (preemptive or non-preemptive)
 - ▣ Earliest Deadline First (EDF), etc.



Scope here: Point-process models
as opposed to fluid-flow queues

M/M/1 queue

- Poisson arrival of packets (first 'M' $\rightarrow$ Markovian) with rate λ
- Exponentially distributed service times of rate μ (second 'M')
- Single Server (1)
- FIFO service discipline
- $\rightarrow Q_t$ = Number of packets in system is continuous-time Markov Process



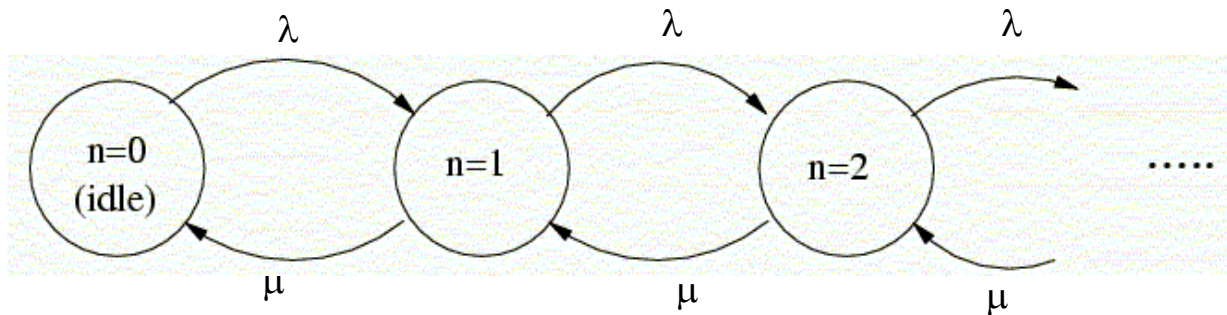
'Derived' Parameter:

- Utilization, $\rho = \lambda / \mu$: if $\rho \geq 1$, instable case (no steady-state q.l.d)

Performance Parameters

- Queue-length distribution: $\pi(t)$, steady-state limit: $\pi = \lim \pi(t)$ (if $\rho < 1$)
- Queue-length that an arriving customer sees
- Waiting/System time distribution
- Buffer-Overflow Probability for level B = Pr(arriving customers sees buffer occupancy B or higher)

M/M/1 queue: Performance



□ Birth-Death Process

- Probability of i packets in queue [using flow-balance equations]

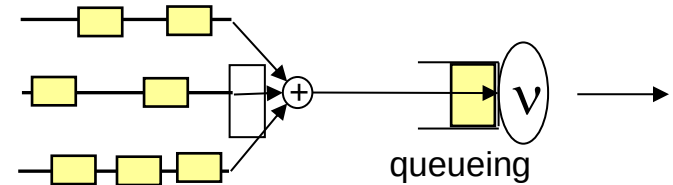
$$\pi_i := \Pr(Q=i) = (1-\rho)^i \rho^i, \text{ where } \rho = \lambda / \mu < 1$$

- Probability of idle server: $\pi_0 = (1-\rho)$
- Average Queue-length: $E\{Q\} = \rho / (1-\rho)$
- Average Delay (System Time): $E\{S\} = E\{Q\} / \lambda = 1 / (\mu - \lambda)$
- Buffer Overflow Probabilities (PASTA principle – Poisson Arrivals See Time Averages)
 $\Pr(Q^{(a)} \geq B) = \Pr(Q \geq B) = \rho^B$

Application : Packet- versus circuit-switched communication

□ Circuit-switched approach

- ISDN and analog telephone NWs, 2G mobile Networks before GPRS
- Allocate resources along the end-to-end path, e.g. allocate one physical channel for one end-to-end call



□ Advantages of Packet-Based Transport

- Flexibility
- Optimal Use of resources of the path, Multiplex-Gain for ,bursty' applications such as sensor data access or web browsing

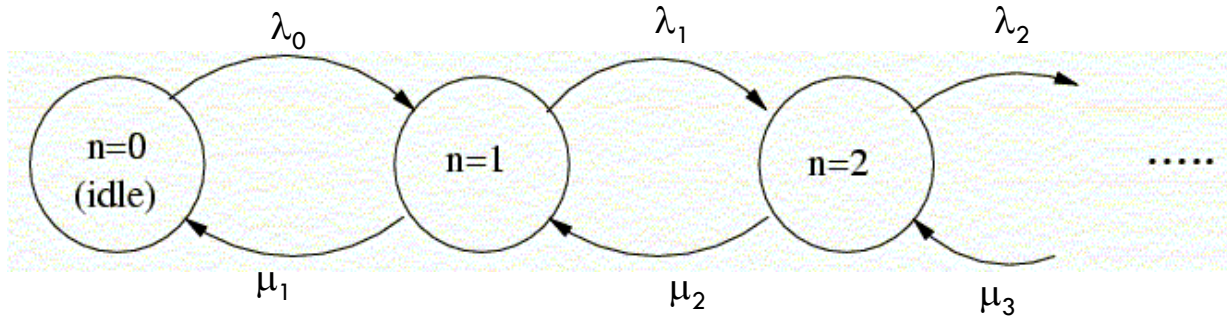
□ Drawbacks

- Buffering/Queueing at routers can be necessary

→ Delay / Jitter / Packet Loss can occur

Models for the analysis of these metrics desired

General Birth-Death Processes



- Steady-State Probabilities (from balance equations):

$$\pi_i := \Pr(Q=i) = \pi_0 \prod_{k=0}^{i-1} \lambda_k / \prod_{k=1}^i \mu_k$$

- Models in this class, e.g.

- M/M/1/B
- M/M/C, M/M/C/C
- Load-dependent services, discouraged arrivals

Exercise 3: multiple server queue, see last slide (15min)

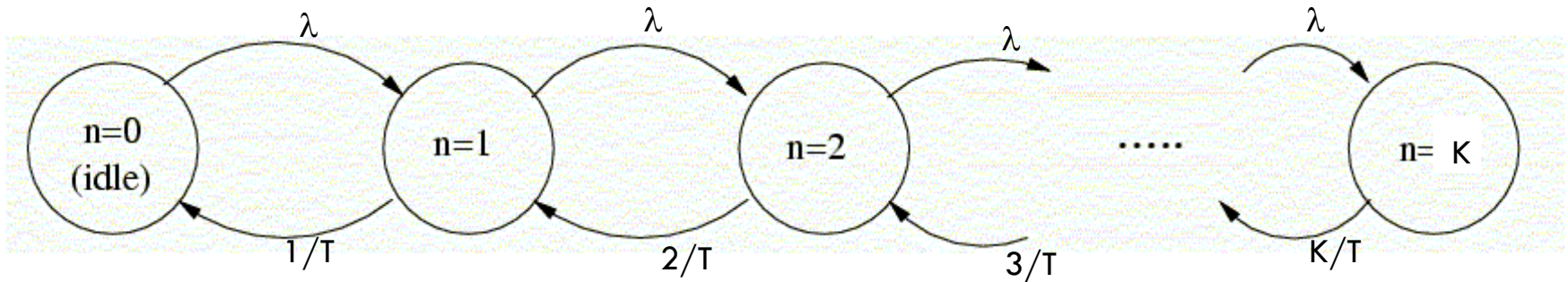
The circuit switched scenario

- K channels
- Users allocate one channel per call for certain call duration
- If all channels are allocated additional starting calls are blocked
- How many channels are necessary to achieve a call certain maximal blocking probability?

Common Model Assumptions:

- Calls are arriving according to a Poission Process (justified for large user population, limit theorems for stochastic processes) with rate λ
- Call durations are exponentially distributed with mean T (okay for voice calls)

Computation of blocking probabilities: M/M/K/K model, Erlang-B formula



□ Finite Birth-Death Process:

- Probability of i calls active

$$\pi_i := \Pr(n=i) = \pi_0 (\lambda T)^i / i! , i=1, \dots, K$$

where $\pi_0 = 1 / [\sum (\lambda T)^i / i!]$ (sum taken over $i=0$ to K)

- Probability of blocked call:

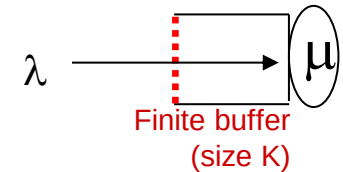
$$P_{(\text{Blocking})} = \pi_K = \pi_0 (\lambda T)^K / K!$$

[also known as Erlang-B formula]

Packet-based link model: M/M/1/K queue

□ Assumptions

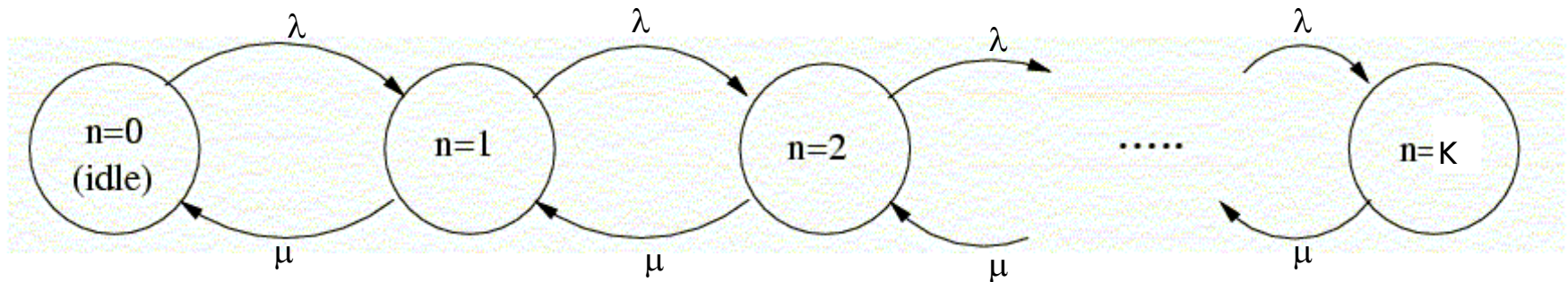
- Poisson arrival of packets with rate λ
- Exponentially distributed service times of rate μ
- Single Server
- Finite waiting room (buffer) for K packets



□ Suitable e.g. for modeling 'bottleneck' link in packet-based wireless networks

- [Full network models: see later in the lecture]

M/M/1/K queue: Performance



□ From Birth-Death Process Theory:

- ▣ Probability of i packets in queue

$$\pi_i := \Pr(Q=i) = (1-\rho)/(1-\rho^{K+1}) * \rho^i, \text{ where } \rho = \lambda/\mu \neq 1, i=0, \dots, K$$

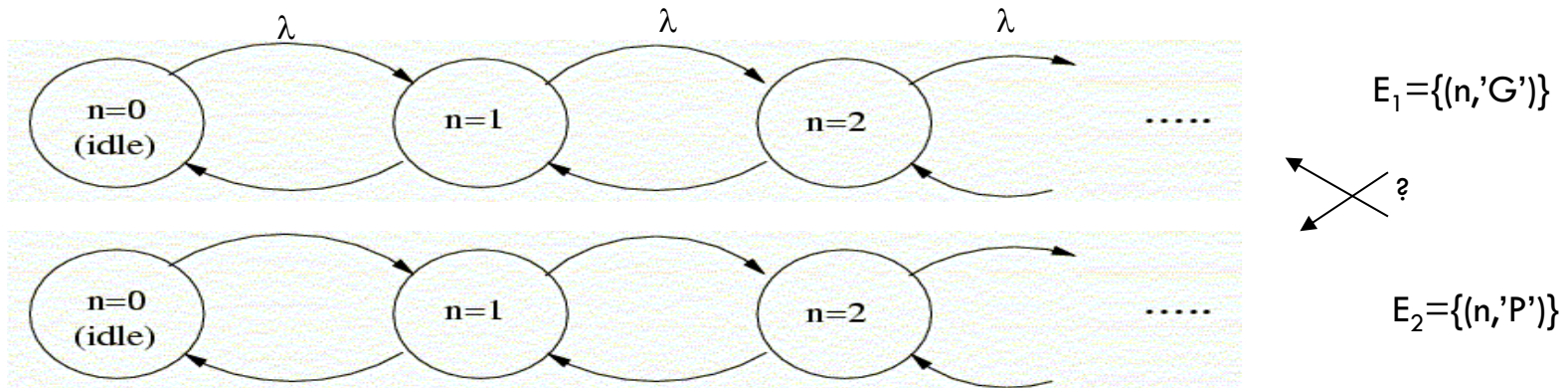
- ▣ Probability of packet loss:

$$p_{(\text{loss})} = \pi_K = (1-\rho)/(1-\rho^{K+1}) * \rho^K$$

- ▣ Average Delay:

$$\check{D} = 1/[\lambda (1-p_K)] * \rho/(1-\rho^{K+1}) * [(1-\rho^K)/(1-\rho) - K \rho^K]$$

Queueing models with varying service



- Extension of $M/M/1$ model for channel-dependent service time, e.g. Gilbert-Elliot model with ARQ
 - ▣ Poor channel condition $\rightarrow$ service rate μ_1
 - ▣ Good channel condition $\rightarrow$ service rate $\mu_2 > \mu_1$
- Two different cases
 - ▣ Channel changes only after successful transition
 - ▣ Channel may change at any moment in time
- Two dimensional Markov chain: $E = \{n=0,1,2,\dots\} \times \{'G', 'P'\}$, Transition Rates?
- Solutions: Truncate and compute π numerically or use Quasi-Birth-Death Process solution

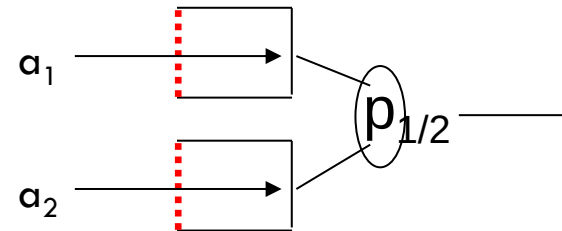
Application to scheduling

Scenario:

- GPRS type of system (TDMA), slotted (discrete time)
- Two users in radio cell:
 - ▣ User 1 with good radio condition, service probability p_1
 - ▣ User 2 worse, service probability $p_2 < p_1$
- Each user has a downstream packet flow, assumed to be Bernoulli arrivals with parameter a_i
- Downstream packets are stored in two different queues

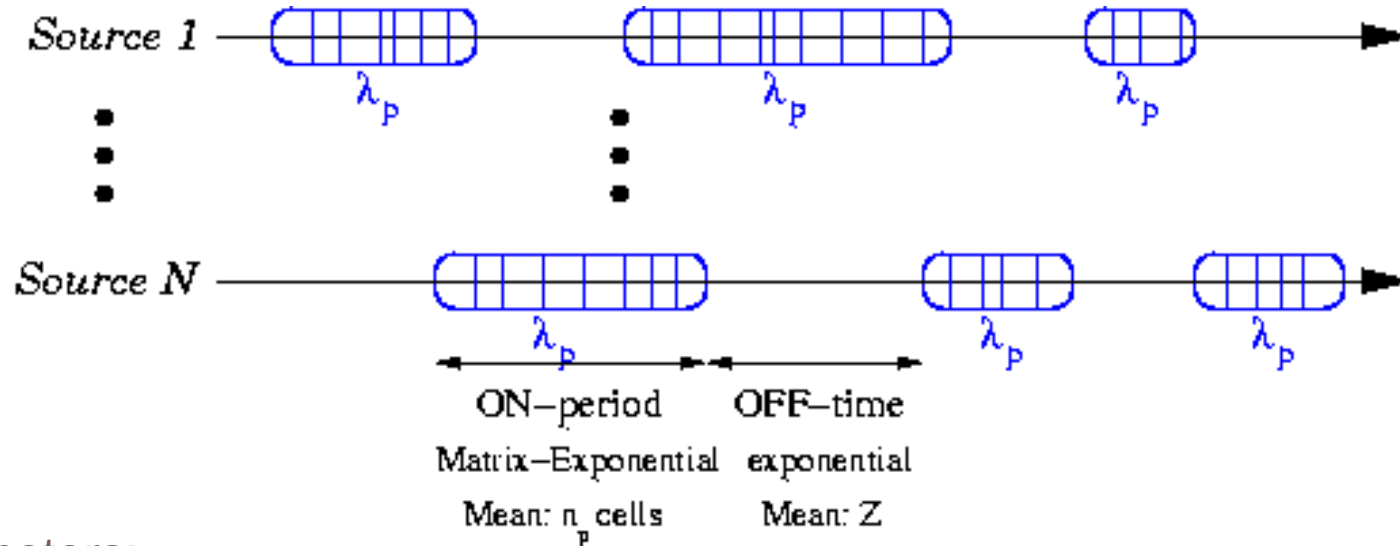
Scheduler needs to make a decision which queue to serve (non-preemptive!)

- ▣ Round-robin
- ▣ Random
- ▣ Strict priority
- Compute buffer occupancy, avg. delay, etc.!
 - ▣ Discrete time Markov chain
 - ▣ State-space: $E = \{(n_1, n_2)\} = \mathbb{N}_0 \times \mathbb{N}_0$
 - ▣ Extended state space (odd/even slot?) for RR
 - ▣ Transition probabilities ?



→ Exercise 5: Analysis of scheduling strategies

More realistic models: ON/OFF Models

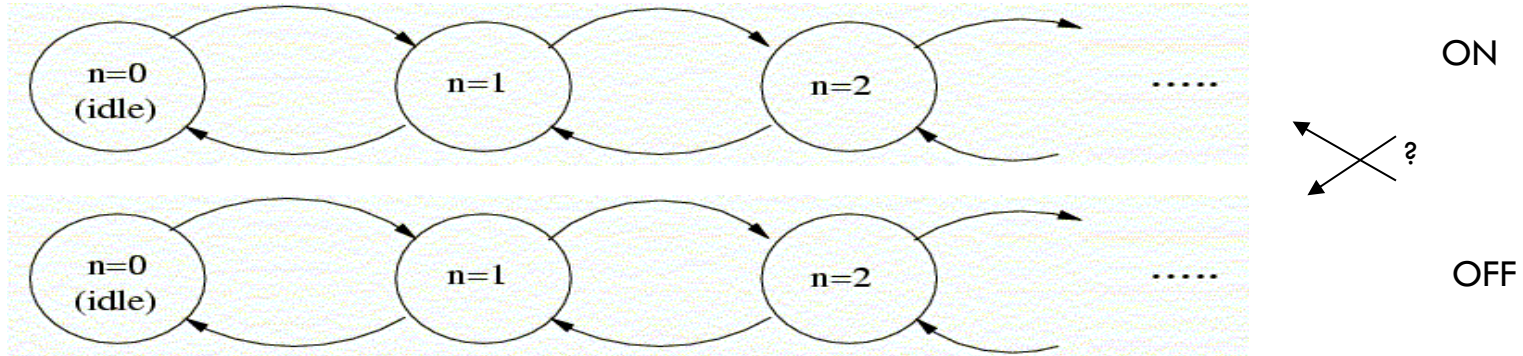


Parameters:

- N sources, each average rate κ
 - During ON periods: peak-rate λ_p
 - Mean duration of ON and OFF times
- } 'bursty' traffic, when $\lambda_p \gg \kappa$

$$\kappa = \lambda_p \text{ ON} / (\text{ON} + \text{OFF})$$

Performance Model: On-OFF/M/1 (single source)



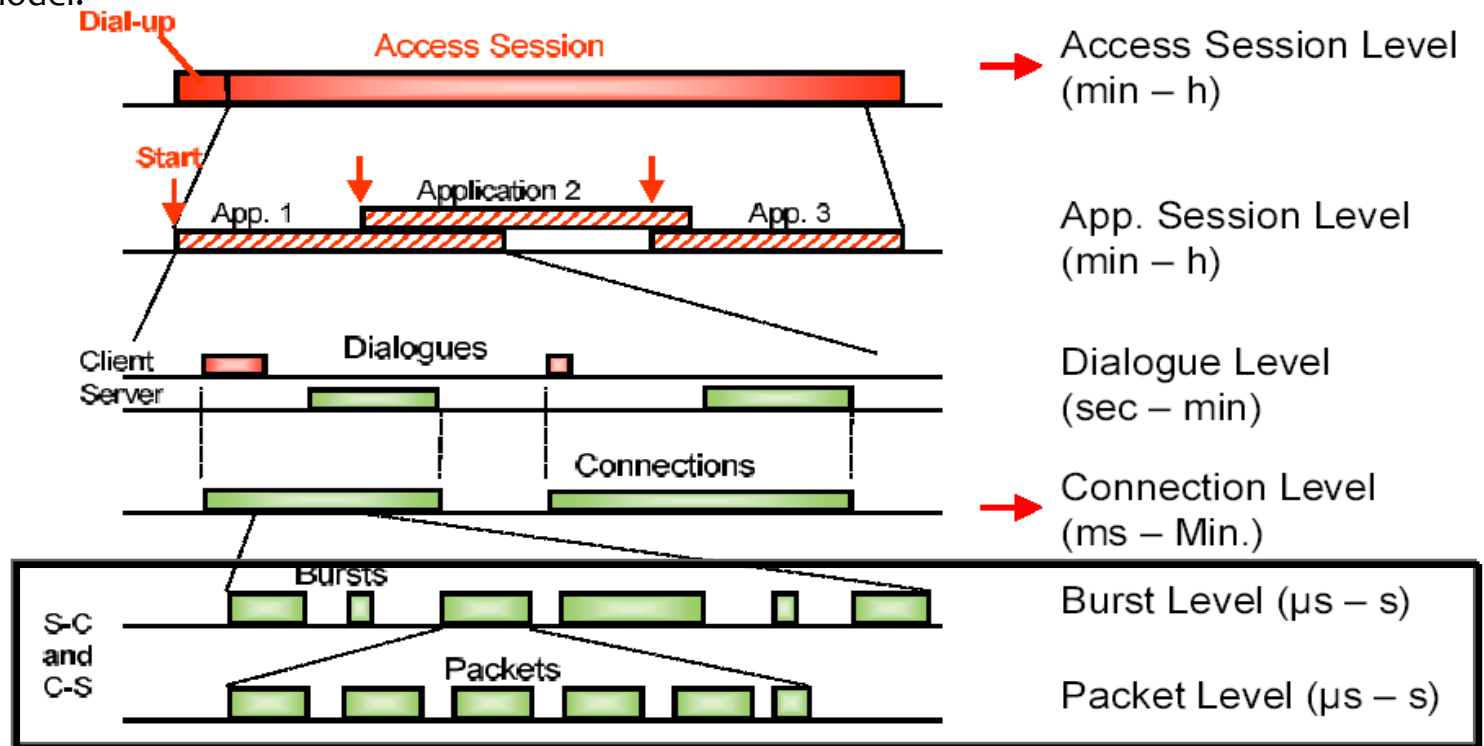
- Two dimensional Markov chain: $E = \{n=0,1,2,\dots\} \times \{\text{'ON'}, \text{'OFF'}\}$
- Transition Rates ?
- Solutions: Truncate and compute π numerically or use Quasi-Birth-Death Process solution

General hierarchical models

□ Frequently used: Several levels with increasing granularity

■ E.g. 3 levels: sessions, connections, packets

■ Or: 5-level model:



Summary

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1) System definitions and examples

- Systems, Systems of Systems, Semester Project scope
- Examples: Wind-farm controller, intelligent distribution grids

2) End-2-end Network Performance

- OSI layer and related performance metrics
- Layer-3 end-to-end performance - delay, loss, throughput

3) Stochastic discrete event models: Markov chains and their application to Layer 2 and Layer 3 performance – Buffering and multiplexing in intermediate nodes

- Continuous Markov models and birth-death processes (revision)
 - M/M/1 queueing models
 - More general models and network traffic characterization
- See C&L Sect 7.3/7.4, 8.1-8.6

Outlook Next lectures

Layer-2 Examples: Field-buses CAN & Flexray

Layer 4 - TCP performance models

Performance assessment of data access schemes

Exercises:

1) Layer 2 delays, see Slide 12

2) throughput, see Slide 14

3) Queueing models

- a) Optional - derive the Chapman Kolmogorov Equation for continuous time Markov chains, see Slide 19
- b) **Multi-server queue:** One router in a network (to be modelled as $M/M/1$ queue) is apparently creating large delays. The admin decides to add a second routing processor to it. In principle, the admin now has the choice to upgrade to an $M/M/2$ system, but he could also configure separate queues for the new routing module (packets will be randomly directed to one of the two queues); alternatively, he could replace the routing processor with another one that is twice as fast. Which solution is performance-wise better? Plot the ratio of the mean queue-lengths for varying arrival rate λ .
- c) **Scheduling (opt):** Implement the Discrete time Markov chain models from Slide 28 (Matlab). Compute the stationary probability vector and the average queue-lengths. Which scheduling strategy is more efficient in the setting of $a_1=a_2=0.2$, $p_1=0.8$, $p_2=0.4$?

References

- Cassandras, Lafortune, '**Introduction to Discrete Event Systems**', Springer, 2021.

Optional additional reading:

- Madsen, JT, Findrik, M, Madsen, TK & Schwefel, H-P 2016, 'Optimizing data access for wind farm control over hierarchical communication networks', *International Journal of Distributed Sensor Networks*, vol. 12, no. 5, 5936235. <https://doi.org/10.1155/2016/5936235>, see <https://vbn.aau.dk/en/publications/optimizing-data-access-for-wind-farm-control-over-hierarchical-co>